

81%

of total flow time was spent in queue.

I built an Excel planning model to examine maintenance delays, forecast incoming workload and compare capacity recovery options.

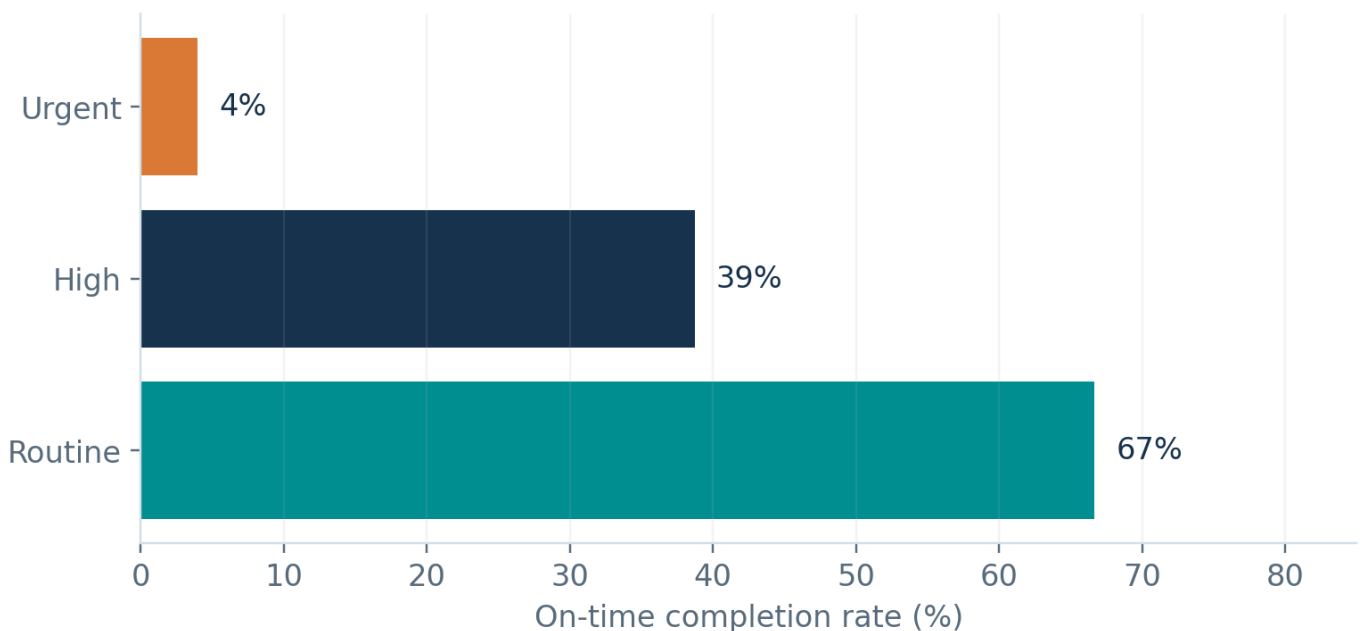
2,400 work packages | 14 aircraft | 2024-2025

The key finding: time before the first labor booking dominates the observed job timeline.

Portfolio simulation using synthetic data. Not affiliated with any aircraft manufacturer.

Only 52% of jobs finished on time.

Late jobs ran 13.5 days late on average. Urgent jobs had the lowest on-time rate in this synthetic dataset.

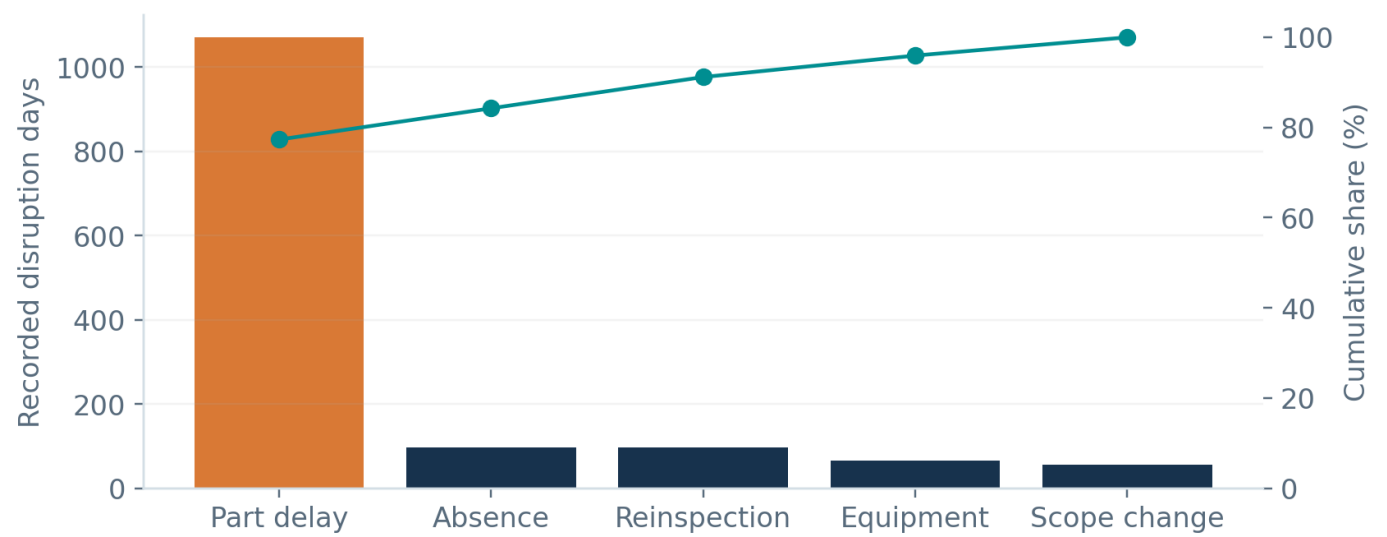


Planning question: How do release timing, due-date allowances and material readiness differ across priorities?

Observed differences identify where to investigate. They do not establish why urgent jobs were late.

Part delays account for 77% of recorded lost days.

Jobs with a recorded part delay were late 76% of the time, versus 47% for jobs with no recorded disruption. Seal kits contributed 232 delay days.

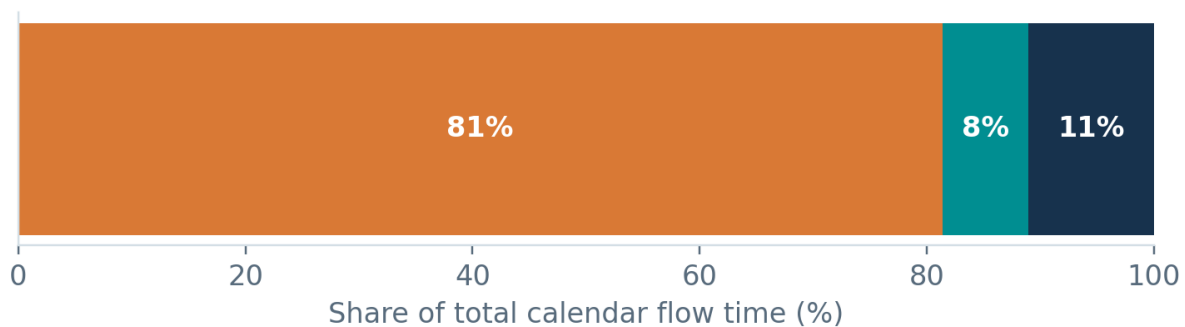


Action to evaluate: Check parts availability before work release, starting with the categories contributing the most recorded delay days.

These are recorded disruption-day totals and associations with lateness, not proof of a complete root-cause explanation.

The largest interval comes before work starts.

I split total calendar flow time into queue, work span and closeout using request, first booking, last booking and completion dates.



Queue 81% | Work span 8% | Closeout 11%

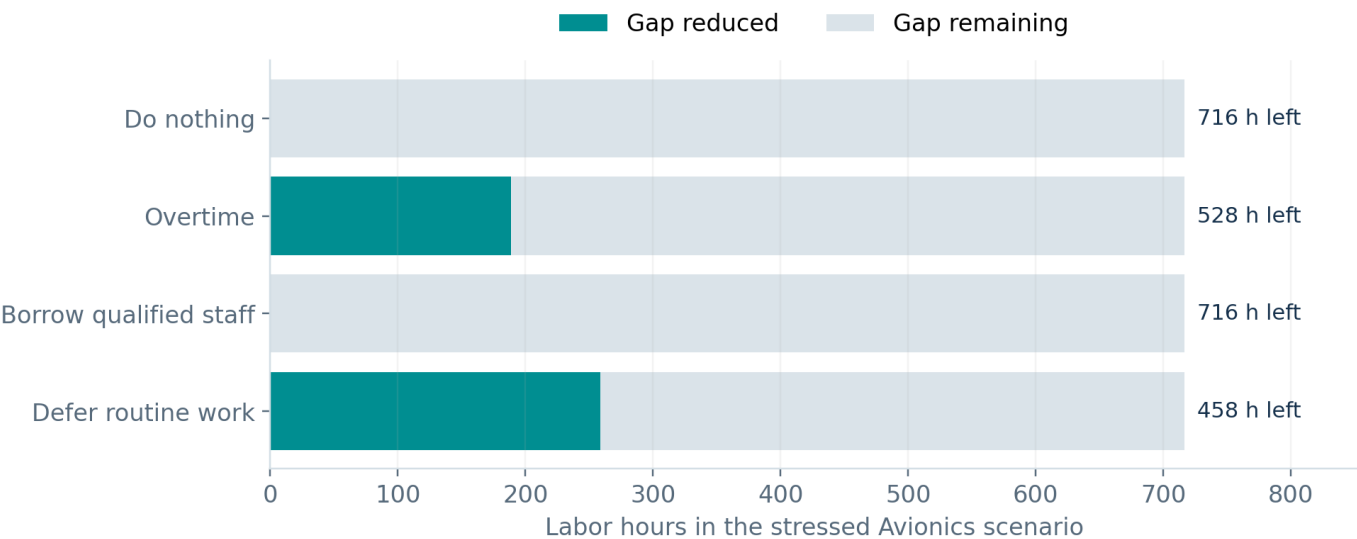
Lean improvement hypothesis

Investigate release readiness and queue constraints before assuming more technician effort will solve the delay. Test whether staging parts reduces waiting.

Work span is elapsed time from first to last booking, not continuous hands-on work. Queue is a timestamp-based proxy; other preparation may occur during it.

A 716-hour gap needs more than one lever.

Stress scenario: Avionics demand increases 25%, plus 120 unplanned hours. No individual modeled option closes the gap.



Overtime: 189 hours recovered for about \$15.6K.
Deferral: 259 hours moved out of the window.
Borrowing: zero available hours under the 85% lending-skill utilization ceiling.

Options are evaluated separately. Deferral is not new capacity. Costs are hypothetical; borrowing requires qualified, available staff. Historical overtime already represents 13% of booked hours.

Holdout bias improved from -7.5% to -0.6%.

The model links work packages, labor bookings, technicians, parts and disruption records to a formula-driven dashboard.

Estimate calibration

Bias moved close to zero, reducing systematic underestimation. MAE also fell 4.1%, from 1.754 to 1.682 hours per job. Task adjustments use only completed work before each cutoff.

Weekly forecast validation

Four methods; 78 initial training weeks, 13 selection weeks and a separate 13-week final test. Selected method: same week last year.

Capacity and scenarios

Forecast incoming estimates plus backlog are compared with skill-specific capacity. Scenario inputs support recovery trade-offs.

Final-test WMAPE: 26.7%. A last-observed-week baseline scored 18.2%, showing that the selection winner was not stable across windows.

Excel tools: structured tables, SUMIFS, INDEX/MATCH, named inputs and scenario dropdowns.